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Hybrid Soft Soil Tire Model (HSSTM). Part I: Tire material and structure modeling

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ABSTRACT

This paper is the first of three papers describing a study focused on developing a lumped-mass discretized tire model using Kelvin-Voigt material elements. The main motivation was to develop a tire model able to capture the dynamic behavior on tire-soft soil interaction, while also being able to run on rigid (flat and rough) surfaces. This paper presents underlying mathematical formulations for characterizing the tire material properties and the kinetics and kinematics of the tire structure; the second paper focuses on the modelling of the tire-terrain interaction. The interface for connecting the tire model developed with the vehicle modeling software CarSim, in order to perform full vehicle simulations is provided.

To minimize the computational time of the code, different techniques were used in stiffness matrix partitioning, parameter initialization, and multi-processing. This resulted in significant improvements for efficiency of the code making it suitable for use by commercially-available vehicle simulation packages.

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1. Introduction

Tire forces and moments depend on the structural behavior of the tire, as well as on terrain behavior. Based on the simulation application (e.g., handling, ride, mobility, durability) and the type of terrain (e.g., deformable, non-deformable, even, uneven) the approach for modeling the tire and the terrain could be different. The tire models used primarily for vehicle simulation on non-deformable terrains, such as FTire (Gipser, 2003), RMOD-K (Oertel, 1997; Oertel and Fandre, 1999), CDTire (Gallrein and Backer, 2007), can be categorized based on usage, accuracy, computational time taken, and degree of parameterization. The number of degrees-of-freedom (DOFs) and consequently the computational effort in these models can range from empirical models (least) to finite element models (largest).

The direct application of an on-road tire model to simulate tire performance on soft soil is not feasible. This is due to the fact that modeling the tire travel on soil requires characterization of stress and strains in the terrain. Moreover, the kinetics and kinematics of the tire on deformable terrains are subjected to different design and operational factors, as well as terrain characteristics. These factors, in addition to the uncertainties that exist in their parameterization, make the formulation of tire-terrain interaction a highly complex problem. Due to this complexity, the number of tire models, similar to the one developed in this research, which are usable in conjunction with multibody dynamic vehicle simulation models, are limited.

The proposed process of developing a complete soft soil tire model was divided into two main sub-processes of mathematical modeling and physical modeling, as shown in Fig. 1. For the mathematical modeling, different components of the system, such as tire material, tire structure, and tire-ground interaction were described using semi-empirical mathematical correlations. Next, these mathematical models were implemented using the MATLAB programming language. The developed code was checked to confirm that the model was correctly implemented and contained no errors. Meanwhile, a physical representation of the problem was essential to provide an insight into the real-world situation. In this regard, an experimental test rig was designed for conducting the related case studies. The type and configuration of these experiments, which were required for validating and parameterizing the implemented sub-models, were developed as an experiment design table which is discussed in (Naranjo, 2013, 2014).

Using the parameters chosen for the computational model, simulations were performed at conditions similar to the experiments. The results from this step were iteratively generated and compared to the test data until the acceptable agreement was achieved. In

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Nomenclature

A	area
A_l	rigid area of a track link as a proportion of its nominal contact area
A_u	parameter characterizing terrain response to repetitive loading
$A(t)$	state matrix
a	half of the contact patch
and	constant terms extracted from the FEM tire model at different tire pressures
B	wheelbase
$B(t)$	input matrix
b	smaller dimension of a rectangular plate or the radius of a circular plate; width
b_{ti}	tire width
c	cohesion
$C_{F\alpha}$	cornering stiffness
C_{Fy}	lateral stiffness
$C(t)$	output matrix
D	diameter
${}_{belt}D^{i,j}$	the relative distance between the center of the jth belt element and its left and right belt element neighbors, described in the ith belt segment local reference frame
$D(t)$	direct transmission matrix
d	diameter
E	modulus of elasticity
e	void ratio; base for the natural logarithm
e_1^i, e_2^i, e_3^i	unit vectors of the local reference frame of the ith belt segment
F	function; thrust, tractive effort
F_x	longitudinal force
F_y	lateral force
F_z	vertical force
F_d	drawbar pull
F_v	tractive effort developed on the vertical shear surfaces on both sides of a track
$F_{pressure}$	normal force to the inner surface of the mass points due to air pressure
H	horizontal component of a tension force
h	thickness tire section height
h_l	lug height
I_{xx}, I_{yy}, I_{zz}	moment of inertia around X, Y, and Z axes
j	shear displacement
κ	slip
$\boldsymbol{k}$	axial material stiffness
K	time-dependent bulk modulus
K_1, K_2	parameters characterizing the shear stress–shear displacement relationship
$K_{measured}$	measured tire stiffness in the radial direction
$K_{modified}$	radial tire stiffness modified based on the pressure change
l	length
l_s	cleat length
M	number of input variables
M_o	limit bending moment per unit length of an ice layer
${}_{sidewall}^{B}M$	sum of individual torque vectors applied by the belt segments to the tire rim
MI	mobility index
n	exponent of sinkage, damping stiffness
n_x, n_y, n_z	Unit vectors parallel to the global X, Y, and Z axes
$n_{segments}$	number of segments
N	number of states
N_{belt_seg}	number of belt segments
N_{seg_elm}	number of elements in each segment
N_{tread_circ}, N_{tread_lat}, N_{tread_total}	number of contact brushes (sensor points) in each belt segment in circumferential and lateral directions, respectively the total number of contact brushes
P	number of output variables
P_d	drawbar pull
p	pressure
${}_{belt}P^i$	point P described at the local reference frame of the ith belt segment
p_{actual}	actual tire inflation pressure
$p_{measured}$	measured tire inflation pressure
$q(t)$	state vector
${}_{rim}^{pos}q$	the position state vector of the wheel
${}_{rim}^{vel}q$	the velocity state vector of the wheel
${}_{rim}q$	the state vector of the wheel
${}_{weg}^{pos}q$	the position state vector of the segment
${}_{seg}^{vel}q$	the velocity state vector of the segment
${}_{seg}q$	the state vector of the segment
${}_{brush}q$	the state vector of a brush element
${}_{total}q$	The state vector of the system
R^2	coefficient of determination
k_c, k_φ	resistance due to terrain compaction
$R_{axis,angle}$	rotation matrix around *axis* with certain *angle*
R_{tot}^i	total transformation matrix for the ith belt segment
RCI	rating cone index
r_{belt}	radius of the tire belt
${}_{seg}^{G}r_{i,j}$	position vector of the jth element in the ith segment in the global reference frame
${}_{seg}^{W}r_{i,j}$	position vector of the jth element in the ith segment in the wheel reference frame
${}_{rim}^{\ G}r$	position vector of the rim in the gloabal reference frame
${}_{cw}^{G}r_i$	position vector of ith belt element from wheel center to wheel in global reference frame
${}_{cw}r_i$	normal position vector of ith belt element
s	shear stress
$s_{\max}$	maximum shear stress
S_{ij}	element compliance
t	time
T	tension
$u(t)$	input vector
$v(t)$	output vector
V_x	longitudinal velocity
V_{sy}	lateral slip velocity
V_j	slip velocity
$width_{tread}$	width tread
W	load, weight
x_{rim}, y_{rim}, and z_{rim}	translational coordinates of the wheel center in the global reference frame
z, z_o	sinkage
α	tire slip angle
α_{rim}, β_{rim}, δ_{rim}	wheel rotation angle around X, Y, Z global axes
γ	tire inclination angle (camber angle)
δ_t	tire deflection
Δy_{seg_elm}	lateral displacement between the centers of two adjacent elements in each tire segment
$\Delta\theta_{seg}$	angular difference between two adjacent belt segments
η	damping stiffness
θ	angle
μ	Poisson's ratio; drawbar coefficient; coefficient of friction
λ	Lagrange multiplier
σ	normal stress
τ	shear stress
ϕ	angle of shearing resistance
ω	angular speed

case the required correlation accuracy was not achieved after extensive simulation iterations, the judgment was made whether to make changes to one of the sub-models, experimental setup, parameterization procedures, or all of the above.

In the following sections, first a brief literature survey for the available tire models that are designed for estimating the tire performance on deformable surfaces is given. Next, the steps that are required for accurately characterizing the tire structural behavior are given in more detail. The main objective of this paper is to present a tire structure model that can describe tire enveloping behavior on non-deformable and deformable surfaces. The choice of tire structure model has a high impact on the tire-transmitted forces, especially in the middle (1–10 Hz) and upper (10–100 Hz) frequency bands of interest (Captain et al., 1979).

It should be stated that the development of the tire-ground interaction model was discussed in a companion paper (Sandu et al., 2019). Next, the developed tire model test bench that accommodates simulating the tire behavior as a standalone module or in conjunction with a vehicle model is discussed. To study tire structural behavior without having a deformable terrain involved, dynamic cleat test simulations were conducted, and results are compared with other benchmarked tire models. Finally, the performance of the HSSTM model in braking and cornering maneuvers on sandy loam terrain in comparison with the experimental results is discussed.

2. Literature survey

As stated earlier, the main challenge in studying the behavior of the vehicle in off-road conditions is characterizing the tire-terrain interaction. Accounting for the tire contact patch changes is essential in estimating tire sinkage into the terrain, and transmitted forces and moments to the vehicle. Historically, a wide variety of models have been developed for formulating and simulating this interaction. The degree of complexity for these models is based on the application, accuracy, and computational cost. Generally, these models can be grouped into three main categories (Taheri et al., 2015): (1) Empirical models, (2) semi-empirical models, and (3) fully physics-based models. The literature survey included in this paper is brief, but the authors have published an extensive review paper on this topic, and the reader is encouraged to refer to Taheri et al. (2015).

Mechanical behavior of the tire and tire-terrain depends on many aspects, such as tire geometrical and material properties, in addition to terrain texture and frictional characteristics (Madsen, 2012). Identifying all of these parameters and correlating them to the vehicle performance using empirical closed-form formulations is limited to similar test conditions. On the other hand, due to the complexity of contact mechanics for deformable objects, using the simple physics-based methods (Crolla et al., 1990) to model the terrain can lead to significant errors, and making the model more complex will increase computational cost (Taheri et al., 2014a). One alternative method for analyzing tire performance is by means of the semi-empirical models (Chan, 2008; Chan and Sandu, 2014a, 2014b, 2014c). In this category of tire models, the tire structure is usually modeled by analytical equations and the terrain is defined using empirically derived models (Wong and Asnani, 2008; Taheri and Wei, 2015). These models are the best nominees for dynamic vehicle simulations because they are a trade-off between accuracy and computational efficiency (Taheri et al., 2014a; Chan, 2008; Chan and Sandu, 2014a).

The majority of the models in this field use the two-dimensional empirical formulation developed by Bekker et al. (1969, 1956), Bekker (1960,1956), Wong and Preston-Thomas (1983), Wong (1984) and Wong and Reece (1967a, 1967b); Wong, 2010). In these formulations, the tire is commonly considered a rigid cylinder, and the normal and shear stresses in the tire contact patch are expressed as functions of the tire kinetics and kinematic variables.

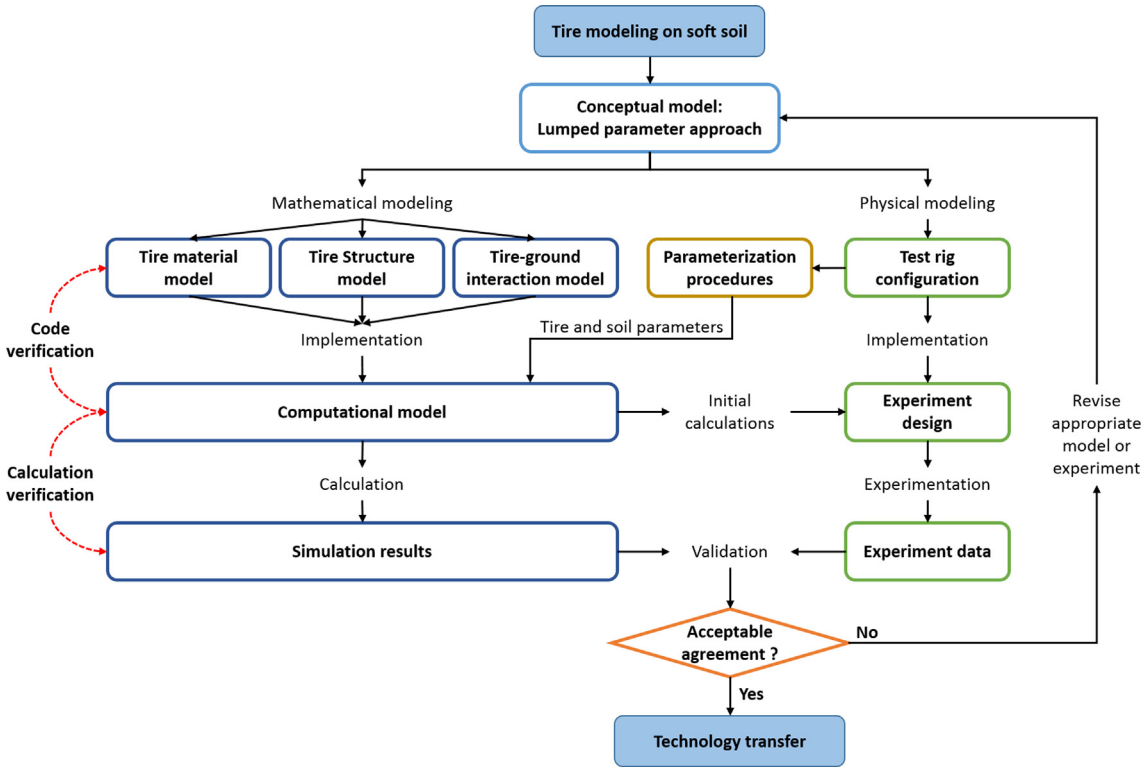


Fig. 1. Semi-empirical mathematical tire modeling on soft soil: Modeling, simulation, and experimental procedures work flow.

Consequently, the corresponding stress components are integrated over the contact patch to calculate the spindle forces, tire sinkage, soil deformation, tire deflection, etc. The more sophisticated tire models use a finite element representation for either tire structure or tire-ground interaction.

The tire model in this study is considered a semi-empirical tire model because it takes a physics-based lumped mass approach for describing the tire structural response in addition to a semi-empirical method for characterizing the tire-terrain interaction.

3. Tire material modeling

A typical modern tire is manufactured from nearly 10–35 different components. Information about tire material properties, processing, mixing, assembly, and curing is almost always confidential, and cannot be obtained from tire companies. Furthermore, the material properties for the same tire from a manufacturer may vary, due to the vulcanization process, for example. In order to accurately estimate the behavior of the tire components, elaborate material models are needed. The parameterization of these models for individual materials in the tire requires performing extensive experimental and analytical procedures, such as elastic and viscoelastic tests on individual tire sections. This level of detail is required for calculating the accurate stress and strains in the tire structure, which would be helpful in the design stage of the tire. For the purpose of the study presented here, simplified methods are chosen for describing the tire material behavior, such as hyperelasticity and viscoelasticity.

3.1. Hyperelasticity

A great portion of the tire structure consists of vulcanized elastomers, such as rubber material. Rubber has a nonlinear and incompressible behavior toward loading, which is independent of the strain rate. This behavior is known as hyperelasticity, and the material which shows this behavior is called green elastic material or hyperelastic. A hyperelastic material differs from an elastic material in four main aspects (Morton, 2013):

- The tire has a high stiffness in the initial step of loading, and dramatically softens in the unloading phase. This phenomenon is known as Mullin's effect.
- Instead of having a hysteresis loop in the stress-strain curves of the loading cycle, the hyperelastic materials have a simple equilibrium curve.
- The hyperelastic materials exhibit different behavior in tension and compression. This is in contrast with Hooke's law, which considers the stress to be proportional to strain. Hyperelastic materials such as rubber, have a higher stress magnitude in compression when compared to the tension for an identical strain magnitude.
- Finally, a hyperelastic material has different modes of deformation that should be studied with respect to the given loading conditions. Each deformation mode requires corresponding constants in the material model that must be characterized experimentally. The choice of model constants and required parameterization tests should be done with care in order to avoid false analytical system response quantities that are not present in the experiments.

The hyperelasticity feature of the rubber should be enhanced with viscoelasticity in order to precisely describe the rate-dependent loading/unloading force-deflection characteristics of the tire.

3.2. Viscoelasticity

Viscoelastic materials show a combined elastic and viscous rate-dependent behavior when experiencing deformation (Taheri et al., 2013). In elastic materials, once the applied stress or strain is removed, the specimen quickly returns to its initial condition. Viscous materials exhibit a resistance toward the shear flow developed due to the applied stress or strain, i.e., upon applying a constant strain, the material creeps. Similarly, by applying a constant stress, the strain increases and then eventually decreases with time.

The internal damping, rolling resistance, and thermal characteristics of a tire are associated with the viscoelastic property of the rubber. Therefore, in order to properly quantify the transient response of the tire, the viscoelastic material property should be incorporated. For small strains, the linear viscoelasticity assumption may be chosen. In this case, the relaxation rate of the material is proportional to the immediate stress, and the total viscoelastic behavior can be expressed using the superposition principle.

3.3. Modeling procedure

There are different mechanical models that can describe the combined hyperelastic and viscoelastic characteristics of a material. Each of the mechanical models considers a certain form of stress or strain response for the material under different loading conditions. The hyperelasticity of the tire was modeled by interpolating the tire load versus deflection data in compression/tension loading/unloading scenarios. Using this approach allows the definition of different loading stiffness for loading and unloading paths.

To incorporate viscoelasticity, three main models were considered including: Maxwell model, Kelvin-Voigt model, and Standard Linear Solid model (Taheri et al., 2014b). For the Maxwell model, the viscoelasticity is modeled using a damping element (Newtonian dashpot) connected to a Hookean spring (stiffness element) in a series configuration. Considering the fact that the Maxwell model exhibits the unrestricted flow of material during loading, it is not desirable for the rubber element modeling. For the Kelvin-Voigt element, the stiffness and the damping elements are connected to each other in a parallel configuration. The force-deflection characteristics of this model for force step input and deflection step input are shown in Fig. 2.

For this type of element, the force-deflection relation under axial loading has the following form:

$$\sigma = k\varepsilon + \eta\frac{d\varepsilon}{dt} \tag{1}$$

where k is the axial stiffness, η is the damping stiffness, ε is the element strain, and σ is the applied stress. It should be noted that stress and strain of the element are analogous to the force and deflection. In the multi-axial loading the equation of motion is written as:

$$S_{ij} = Ke_{ij} + \eta\frac{de_{ij}}{dt} \tag{2}$$

In which K is the time-dependent bulk modulus, and S_{ij} is the element compliance. Another material model of interest is the Standard Linear Model (SLM), which is a Maxwell model that is connected to another stiffness element in a parallel configuration. It is cumbersome to solve for the stress value in the Maxwell arm, since it contains both the stress and its derivative. Using the SLM model resulted in less than 1% improvement in estimating the tire deflection, which was insignificant and indefensible considering the added computational cost. Therefore, the Kevin-Voigt element

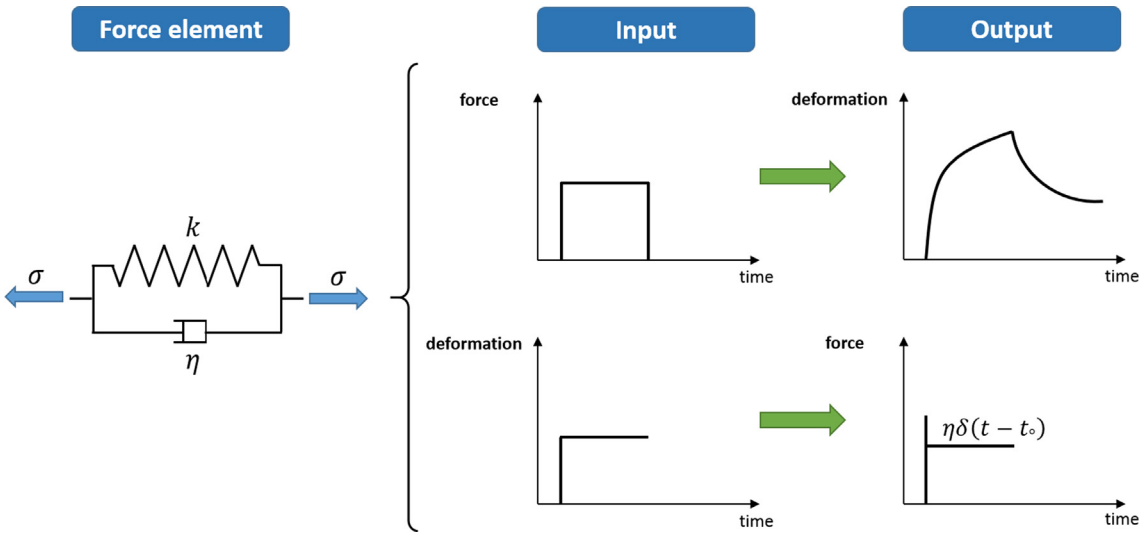


Fig. 2. The force-deflection characteristics of the Kevin-Voigt model for force step input and deflection step input.

was used as the main force element between tire/rim elements due to the fact that it is more accurate compared to the Maxwell model, and easier to solve for the stress values compared to the SLM.

4. Tire structure modeling

As discussed earlier, the detailed modeling of the tire structure is not required for studying the mobility of the tire. This is due to the fact that, for evaluating the tire performance, only a limited number of parameters are needed, such as forces and moments at the spindle, and wheel sinkage. Therefore, modeling the tire with a coarse level of tire structure discretization would be adequate, and can result in a fast computational time. This feature is essential for full vehicle simulations and control applications when real-time simulation is required. The lumped parameter models reduce the DOFs in the model in favor of the computational effort, and consider the simplified material models in the respective directions. Such a method was used in HSSTM for representing the tire structure.

In the early phase of the project, a simplified lumped mass approach for modeling the tire structure was introduced by Pinto (2012) and Sandu et al. (2012, 2011) This approach considered the tire structure as three layers of lumped masses, in which the masses are connected to the rim and also to each other through a set of linear springs and dampers. This three-layer structure approach was an advanced version of a lumped mass single layer on-road tire model developed by Umsrithong and Sandu (2010a, 2010b, 2012a, 2009, 2012b) and Umsrithong (2012). In 2012, an advanced method for modeling the tire was introduced (Taheri, et al., 2013), and a more systematic approach was used for developing the software. In this new approach, the tire belt is discretized circumferentially in multiple belt segments that are suspended on the rim using Kelvin-Voigt elements, which include variable stiffness and damping. These nonlinear elements capture the effect of the temperature and pressure changes on the tire mechanical characteristics through a set of empirical equations. Each belt segment is divided into a series of lumped masses connected to each other with in-plane and out-of-plane springs and dampers. The dynamics of these lumped masses, in addition to the wheel mass, is described in a state-space representation. The state is a set of variables that, along with the time step, characterize the individual configuration of the system at any instance of time. The state variables are defined by equations of motion, and can be positions, velocities, acceleration, force, moment, torque, pressure, etc. The state variables that are described using the differential equations are called state differential variables, and those that are defined directly from dynamic conditions, are called extra state variables. The standard notational convention (Kim and Nelson, 1999) for describing a state-space representation is as follows:

$$\text{State equation}: \quad \dot{q}(t)_{[N\times 1]} = A(t)_{[N\times N]}q(t)_{[N\times 1]} + B(t)_{[N\times M]}u(t)_{[M\times 1]} \tag{3}$$

$$\text{Output equation}: \quad v(t)_{[P\times 1]} = C(t)_{[P\times N]}q(t)_{[N\times 1]} + D(t)_{[P\times M]}u(t)_{[M\times 1]} \tag{4}$$

where $q(t)$ is the state vector, $\dot{q}(t)$ is the derivative of the state vector, $A(t)$ is the state matrix, $B(t)$ is the input matrix, $C(t)$ is the output matrix, $D(t)$ is the direct transmission matrix, $u(t)$ is the input vector, $v(t)$ is the output vector, N is the number of states, M is the number of input variables, and P is the number of output variables. It should be noted that the input, output, and state vectors, as well as all the state-space representation matrices are time dependent. The choice of the state variables for different sections of the

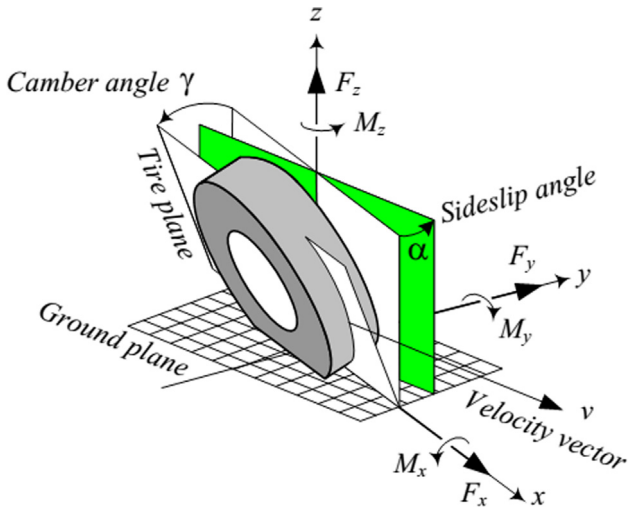


Fig. 3. The representation of the C-axis coordinate system.

tire model is not unique, and would be discussed accordingly in the following sections. The type of mathematical model used to represent the tire structure is called a tire realization.

After discretizing the tire into smaller elements, dynamics of each element can be expressed using a set of first order and second order differential equations. The second order differential equations can be rearranged as a set of first order ODEs. The complete set of the ODEs can be shown as follows:

$$\left\{\begin{array}{c} \dot{q}_1 = f_1(x_1, ..., x_N, u_1, ..., u_M, t) \\ \vdots \\ \dot{q}_N = f_N(x_1, ..., x_N, u_1, ..., u_M, t) \end{array}\right\} \tag{5}$$

$$\left\{\begin{array}{c} \dot{v}_1 = g_1(x_1, ..., x_N, u_1, ..., u_M, t) \\ \vdots \\ \dot{v}_N = g_N(x_1, ..., x_N, u_1, ..., u_M, t) \end{array}\right\} \tag{6}$$

where $f_i(i = 1, ..., N)$, and $f_j(j = 1, ..., N)$, $g_j(j = 1 \cdots P)$ include the following: (1) nonlinear functions of states and/or inputs, such as *Sin* and *Cos* functions, (2) terms with states and/or inputs appearing as powers of something other than 1 and 0, an (3) terms with cross products of states and/or inputs. As a result, the multi-segments model that represent the tire characteristics is an autonomous (time-variant) non-linear system.

4.1. Coordinate system convention

Before defining the state variables of the system, the coordinate system for the sign convention was defined. The definitions for the coordinate systems used in this study are similar to the Tyre Data Exchange format (TYDEX). TYDEX is a conventional interface between tire measurements and tire models developed and unified by an international tire working group to make the tire measurement data exchange easier. Additionally, TYDEX introduce an interface between the tire model and simulation tool called Standard Tire Interface (STI) (Oosten, 1997; Van Oosten, 1999), which was adopted in the model interface (described in detail in Section 7) with the vehicle model.

Along with the global reference frame, one additional right-hand orthogonal axis system used is the C-axis system (center axis system), as shown in Fig. 3. The angles of rotation illustrated in this figure are: a positive slip angle α, a positive inclination angle γ, and a positive wheel rotation speed ω. The C-axis coordinate origin is mounted at the center of the wheel rim. The X_c axis is in the central wheel plane and is parallel to the ground. The central wheel plane is constructed by decreasing the width of the wheel until it becomes a rigid disk with zero width. The Y_c axis is the same as the spin axis of the wheel and rotates with the inclination angle γ. The Z_c axis is in the central plane of the wheel, points upwards, and turns with the inclination angle γ (camber).

4.2. Wheel system

The rim kinematics can be described using six degrees of freedom (DOF) resulting in 12 state variables. Consider the position state vector of the wheel system as following:

$${}^{pos}_{rim}q = [x_{rim}, y_{rim}, z_{rim}, \alpha_{rim}, \beta_{rim}, \delta_{rim}]^T \tag{7}$$

where x_{rim}, y_{rim}, and z_{rim} are the translational coordinates of the wheel center in the global reference frame, α_{rim} is the wheel rotation angle around global X axis, β_{rim} is the wheel rotation angle around the global Y axis, and δ_{rim} is the wheel rotation angle around the Z axis. Furthermore, the velocity state vector of the wheel system is:

$${}^{vel}_{rim}q = [\dot{x}_{rim}, \dot{y}_{rim}, \dot{z}_{rim}, \omega_x, \omega_y, \omega_z]^T \tag{8}$$

where $\dot{x}_{rim}, \dot{y}_{rim}$, and $\dot{z}_{rim}$, and are rim center translational velocities along global X, Y, and Z axes described in the global reference frame. ω_x, ω_y, and ω_z are also rim center rotational velocities around global X, Y, and Z axes described in the global reference frame. Therefore, the final state vector of the wheel is given as:

$${}_{rim}q = \begin{bmatrix} {}^{pos}_{rim}q \\ {}^{vel}_{rim}q \end{bmatrix} \tag{9}$$

4.3. Tire belt

By discretizing the tire belt circumference into N_{belt_seg} segments, the angle between the centers of each two segment will be

$$\Delta\theta_{seg} = \frac{2\pi}{N_{belt_seg}} \tag{10}$$

Next, each segment is divided into N_{seg_elm} segment elements. The number of belt elements is assumed to be an odd number greater than three in order to always have at least one node at the middle of tire width and two neighboring nodes. Each segment element is actually a lumped mass with only translational DOF. Eliminating the rotational DOF from the belt segments helps reduce the computational effort of the model, while maintaining its accuracy. Consequently, the state vector for the segment elements is written as:

$${}_{seg}q = \begin{bmatrix} {}^{pos}_{seg}q_i \\ {}^{vel}_{seg}q_i \end{bmatrix} i = 1, ..., N_{belt_seg} \tag{11}$$

where

$${}^{pos}_{seg}q = [x_j, y_j, z_j]^T_i, j = 1, \cdots, N_{seg_elm} \tag{12}$$

$${}^{pos}_{vel}q = [\dot{x}_j, \dot{y}_j, \dot{z}_j]^T_i, j = 1, \cdots, N_{seg_elm} \tag{13}$$

4.4. Discretization level

The number of segment elements depends on the smallest wavelength of the road perturbations. This concept is shown in

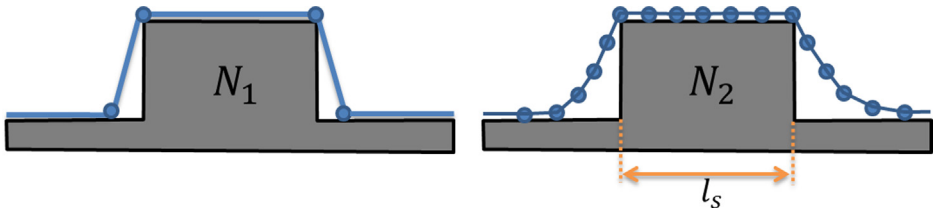


Fig. 4. The discretization level effect on simulating the road perturbation.

Fig. 4. When the tire has only two elements on the cleat the tire deformation and consequently structural forces are not accurate.

The relation between the road wavelength (or cleat length l_s in this case), and number of the belt segment on a tire with radius R can be expressed as (Oertel, 1997):

$$N_{belt_seg} \geq \frac{20\pi R}{l_s} \tag{14}$$

By increasing the number of belt segments, the accuracy of model in enveloping over the road disturbances increases. At the same time, the computational cost increases due to the added number of differential equations associated with each lumped mass. Because an explicit integration method was used, the solver step size should be chosen based on the highest frequency in the system to maintain numerical stability. An increasing number of belt segments, due to short road wavelength inputs, leads to higher frequencies and smaller solver step size. This adds to the computational load of the model. Having these limitations in mind, the number of tire belts segments were chosen between 70 and 140 elements, that corresponds to an acceptable minimum road wavelength range of 0.35–0.17 (m) respectively for the tire used in this study (Michelin LTX radius: 0.3929 m). Also, the number of belt elements in each belt segment (lateral belt discretization) was chosen between 3 and 6 elements.

4.5. Pressure effect

The general behavior of the tire depends substantially on tire inflation pressure. For many tire types, as the tire is loaded, its stiffness increases non-linearly. Meanwhile, there is a constant term in the load-deflection curves for the tire mass elements due to the inner pressure force. The air pressure results in a force normal to the inner surface on each of the mass points. This force is calculated using the following formula:

$$F_{pressure} = \frac{\pi \cdot r_{belt}}{n_{segments}} \cdot width_{tread} \cdot P_{actual} \tag{15}$$

Increasing the tire pressure causes the tire structure to become stiffer, which makes the contact patch to shrink at constant normal load. The small contact patch increases the tire sinkage into the terrain that increases the soil bulldozing force on the tire.

4.6. Stiffness changes

Tire stiffness varies by different factors such as inflation pressure, rolling speed, tire size, and tire age (Lines and Murphy, 1991). To include the pressure change effects on the tire structure, the tire stiffness in the radial direction is updated at each time step based on the tire inflation pressure:

$$K_{modified} = \left(a_1 + a_2 \cdot \frac{p_{actual}}{p_{measured}}\right) \cdot K_{measured} \tag{16}$$

where a_1 and a_2 are constant terms, which are specified based on curve fitting developed finite element tire model simulation results at different tire pressures. This procedure results in a look-up table, which is used for interpolating the relative stiffness of the tire in the radial direction and at every normal load.

The tire stiffness nonlinear change with speed is mostly due centrifugal stresses in the tire structure. The stiffness of a rolling tire decreases significantly at low rolling speeds, and at speeds above 10 km/h is relatively constant. The tire speed influence on the structural stiffness was not included in the model, because sufficient data for studying this effect at different tire speeds was not available.

Furthermore, tire damping varies with different factors that in most cases are highly correlated. The following factors play significant role in tire damping variations (Lines and Murphy, 1991): tire age, road roughness, and tire inflation pressure. Tire rolling speed, tire load, vibration amplitude, vibration frequency, and driving torque also affect the tire damping. The variety of effective factors, their correlation, and lack of appropriate methods and instruments for measuring the tire damping (at loading conditions) make it challenging to model tire damping changes within the scope of this study.

4.7. Tire tread

The Michelin LTX tire in this study was tested both with and without surface tread. In order to incorporate the tread design, a certain number of brush elements are assigned to each belt element in a rectangular array. Consider an array of bristles with N_{tread_circ} elements in the circumferential direction and N_{tread_lat} elements in the lateral direction. As a result, the number of total sensor points assigned to each belt element will be:

$$N_{tread_total} = N_{tread_circ} \times N_{tread_lat} \tag{17}$$

Each brush element is a massless bristle that has translational stiffness in radial, longitudinal, and lateral directions. The base of these bristles is connected to each lumped mass (element segment), and the tip is touching the ground. This massless tip acts as a sensor point and can be used to detect the tire-road contact. Using the direction and value of the deflection in the bristle, the ground forces generated are also calculated. Through implementation of the sensor points, it is possible to increase the resolution of the contact patch by having more contact detection points at this area. At the same time, to define their velocities in space, these massless points only require extrapolation of the position and velocity of the nearby belt elements without the need for differential equations. Based on the findings of (Sandu et al., 2019), there are only three state variables needed to describe the position of the brush tips in the space. These variables store the displacements of the brush tips in the global reference frame from previous iterations of the solver. Therefore, the state vector of each brush element will be:

$${}_{brush}q_{i,j}^{m,n} = \begin{bmatrix} \lambda_{i,j}^{(1)} \\ \lambda_{i,j}^{(2)} \\ \lambda_{i,j}^{(3)} \end{bmatrix} \quad \begin{cases} i = 1, \ldots, N_{tread_lat} \\ j = 1, \ldots, N_{tread_circ} \end{cases} \tag{18}$$

where m and n are the indices for the associated n^{th} belt element in the mth belt segment. The final state vector of the system can be constructed as follows:

$${}_{total}q = \left[{}_{rim}q, {}_{seg}q, {}_{brush}q, \right]^T \tag{19}$$

4.8. Initial position initialization

The initial position of the rim is expressed in the global reference frame by:

$${}_{rim}^{G}r_0 \tag{20}$$

The subscript to the left of the symbol represents the element for which the position vector is defined. The superscript at the left of the position vector represents the reference frame with respect to which the position vector is defined (G is used for the global reference frame, W is used for the wheel local reference frame). The initial position of the tire elements depends on the tire geometrical properties, as well as the rim initial camber and slip angles. Consider the tire element P, with the position vector ${}^{W}r$ described in the wheel local reference frame noted as C-Axis. If this reference

frame rotates around global X, Y, and Z axes respectively, the coordinates of the point P in the global and in the rim local reference frames can be related using the equation:

$$^{G}r = R^{W}_{total} r \tag{21}$$

where R_{total} is the total transformation matrix after three consecutive rotations and can be calculated by multiplying individual transformation matrices for the rotation around each axis respectively:

$$R_{total} = R_{Y,\theta} R_{X,\psi} R_{Z,\phi} \tag{22}$$

The position vector of the jth element in the first tire segment is:

$$^{W}_{seg}r_{1,j} = \begin{bmatrix} R \\ \left(j - \frac{N_{seg_elm}+1}{2}\right)\Delta y_{se_elm} \\ 0 \end{bmatrix} \tag{23}$$

where Δy_{seg_elm} is the lateral displacement between the centers of two adjacent elements in each tire segment:

$$\Delta y_{seg_elm} = \frac{tire_width}{N_{seg_elm} - 1} \tag{24}$$

Consequently, the local position of the jth element in the ith segment is:

$$^{W}_{seg}r_{i,j} = R_{Y,\left(\Delta\theta_{seg}\times i\right)} \times {}^{W}_{seg}r_{1,j} \tag{25}$$

where $\Delta\theta_{seg}$ is the angular difference between two adjacent belt segments:

$$\Delta\theta_{seg} = \frac{2\pi}{N_{belt_seg}} \tag{26}$$

The initial position of the jth element in the ith belt segment described in the global reference frame can be expressed as:

$$^{G}_{seg}r_{i,j} = {}^{G}_{rim}r_o + R_{Z,(Slip_angle)} \times R_{Z,(Camber_angle)} \times {}^{W}_{seg}r_{i,j} \tag{27}$$

Finally, the initial state vector of the system becomes:

$$_{total}q_{\circ} = \begin{bmatrix} {}_{rim}q_{\circ} \\ {}_{seg}q_{\circ} \\ {}_{brush}q_{\circ} \end{bmatrix} \tag{28}$$

4.9. Tire model kinematics

In order to write the equations of motion for the rim and the lumped masses, the internal forces between the tire elements and the rim should be identified. The Kelvin-Voigt force elements connect tire belt elements to each other and to the rim. Their force generation is a function of relative displacement and relative velocity of lumped masses with respect to the rim circumference. Therefore, formulating the model kinematics is essential for calculating the kinetics of the elements. In this section, a set of kinematic parameters that are required for writing the model equations of motion are introduced.

The transformation matrix for the ith belt segment is defined as:

$$R^{i}_{tot} = R_{Z,\phi_{rim}} \times R_{X,\psi_{rim}} \times R_{Y,\left(\theta_{rim}+\Delta\theta_{seg}\times i\right)} \tag{29}$$

For the un-inflated tire, the unity vector $_{cw}Gr_i$ normal to the center of the ith belt segment, which passes through the rim center is given as:

$$^{G}_{cw}r_i = R^{i}_{tot} \times \begin{bmatrix} R \\ 0 \\ 0 \end{bmatrix} \tag{30}$$

$$^{G}_{cw}\widehat{\boldsymbol{r}}_i = \frac{^{G}_{cw}\boldsymbol{r}_i}{\left|^{G}_{cw}\boldsymbol{r}_i\right|} \tag{31}$$

The caret superscript (^) indicates a vector normalized by its length. The vector from the rim center to the jth element of the ith belt segment of the tire is:

$$^{G}_{cwo}r_{i,j} = R^{i}_{tot} \begin{bmatrix} R \\ \left(j - \frac{N_{seg_elm}+1}{2}\right)\Delta y_{seg_elm} \\ 0 \end{bmatrix} \tag{32}$$

$$^{G}_{cwo}\widehat{\boldsymbol{r}}_{i,j} = \frac{^{G}_{cwo}\boldsymbol{r}_{i,j}}{\left|^{G}_{cwo}\boldsymbol{r}_{i,j}\right|} \tag{33}$$

The relative position of the first belt element near the left sidewall in the ith belt segment from its projection on the rim is given as:

$$^{G}_{wtl}r_i = {}^{G}_{tire}r_{i,N_{belt_seg}} - {}^{G}_{rim}r_{i,N_{belt_seg}} \tag{34}$$

Similarly, the relative position of the last belt element near the right sidewall in the ith belt segment from its projection on the rim is given as:

$$^{G}_{wtr}r_i = {}^{G}_{tire}r_{i,1} - {}^{G}_{rim}r_{i,1} \tag{35}$$

For the intermediate belt elements located between the first and last elements, which are directly connected to the rim, the relative position of each element from the center of its left and right neighbor elements are:

$$^{G}_{left_elm}r_{i,j} = {}^{G}_{tire}r_{i,j} - {}^{G}_{tire}r_{i,j+1} \quad j = 2, \ldots, N_{seg_elm} - 1 \tag{36}$$

$$^{G}_{right_elm}r_{i,j} = {}^{G}_{tire}r_{i,j} - {}^{G}_{tire}r_{i,j-1} \quad j = 2, \ldots, N_{seg_elm} - 1 \tag{37}$$

The relative position of the tread block center from its neighbor j^{th} belt element is:

$$^{G}_{tread}r_{i,j} = {}^{G}_{tire}r_{i,j} + |tread_tickness| \times {}_{cw}\hat{r}_i \tag{38}$$

The local coordinate system at the center of the ith belt segment described in global reference frame is:

$$^{G}_{tread}r_{i,j} = {}^{G}_{tire}r_{i,j} + |tread_tickness| \times {}_{cw}\hat{r}_i \tag{39}$$

The position of the point P described at the local reference frame of the ith belt segment is:

$$_{belt}P^{i} = p_1 \cdot e^{i}_1 + p_2 \cdot e^{i}_2 + p_3 \cdot e^{i}_3 \tag{40}$$

where

$$e^{i}_1 = \frac{\widetilde{e}^{i}_2 \times {}_{cw}\hat{r}_i}{\| \widetilde{e}^{i}_2 \times {}_{cw}\hat{r}_i \|}, \quad e^{i}_2 = \frac{R_{Z,\varphi_{rim}} \times R_{X,\psi_{rim}} \times [0 \quad 1 \quad 0]^T}{\| R_{Z,\varphi_{rim}} \times R_{X,\psi_{rim}} \times [0 \quad 1 \quad 0]^T \|}, \quad e^{i}_3 = {}_{cw}\hat{r}_i \tag{41}$$

$$\widetilde{e}^{i}_2 = \begin{bmatrix} 0 & -e^{1}_{2z} & e^{i}_{2y} \\ e^{i}_{2z} & 0 & -e^{i}_{2x} \\ -e^{i}_{2y} & e^{i}_{2x} & 0 \end{bmatrix} \tag{42}$$

The tire structural stiffness and damping behaviors are simulated using the tire model Kelvin-Voigt elements. The force that is produced by a Kelvin-Voigt element is proportional to its length and the relative velocity between its two ends. The direction of this force is parallel to the element centerline. Therefore, the position and velocity of the force element tips relative to their bases should be calculated.

The relative distance between a lumped mass in the jth belt element and its projected position on the rim, described in the ith belt segment local reference frame is expressed as:

$$ {}_{belt}D^{i,j} = DT_1^{i,j} \cdot e_1^i + DT_2^{i,j} \cdot e_2^i + DT_3^{i,j} \cdot e_3^i \tag{43} $$

where

$$ \begin{bmatrix} DT_1 \\ DT_2 \\ DT_3 \end{bmatrix}^{i,N_{belt_elm}} = \begin{bmatrix} e_1^i \\ e_2^i \\ e_3^i \end{bmatrix}^T \cdot {}^G_{wtl}r \tag{44} $$

$$ \begin{bmatrix} DT_1 \\ DT_2 \\ DT_3 \end{bmatrix}^{i,1} = \begin{bmatrix} e_1^i \\ e_2^i \\ e_3^i \end{bmatrix}^T \cdot {}^G_{wtr}r \tag{45} $$

The relative distance between the center of the jth belt element and its left and right belt element neighbors, described in the ith belt segment local reference frame is given as:

$$ \begin{bmatrix} DBL_1 \\ DBL_2 \\ DBL_3 \end{bmatrix}^{i,j} = \begin{bmatrix} e_1^i \\ e_2^i \\ e_3^i \end{bmatrix}^T \cdot {}_{left_elm}{}^G r_{i,j} \quad j = 1, ..., N_{seg_elm} - 1 \tag{46} $$

$$ \begin{bmatrix} DBR_1 \\ DBR_2 \\ DBR_3 \end{bmatrix}^{i,j} = \begin{bmatrix} e_1^i \\ e_2^i \\ e_3^i \end{bmatrix}^T \cdot {}^G_{wtl}r - {}_{right_elm}{}^G r_{i,j} \quad j = 2, ..., N_{seg_elm} \tag{47} $$

Relative velocity measurements

$$ {}^W DTD = \dot{DT}_1^{i,j} \cdot e_1^i + \dot{DT}_2^{i,j} \cdot e_2^i + \dot{DT}_3^{i,j} \cdot e_3^i \tag{48} $$

$$ \begin{bmatrix} \dot{DT}_1 \\ \dot{DT}_2 \\ \dot{DT}_3 \end{bmatrix}^{i,N_{belt_elm}} = \begin{bmatrix} e_1^i \\ e_2^i \\ e_3^i \end{bmatrix}^T \cdot {}^G_{wtl}\dot{r}^i \tag{49} $$

$$ \begin{bmatrix} \dot{DT}_1 \\ \dot{DT}_2 \\ \dot{DT}_3 \end{bmatrix}^{i,1} = \begin{bmatrix} e_1^i \\ e_2^i \\ e_3^i \end{bmatrix}^T \cdot {}^G_{wtr}\dot{r}^i \tag{50} $$

$$ \begin{bmatrix} \dot{DBL}_1 \\ \dot{DBL}_2 \\ \dot{DBL}_3 \end{bmatrix}^{i,j} = \begin{bmatrix} e_1^i \\ e_2^i \\ e_3^i \end{bmatrix}^T \cdot {}_{left_elm}{}^G\dot{r}_{i,j} j = 1, ..., N_{seg_elm} - 1 \tag{51} $$

$$ \begin{bmatrix} \dot{DBR}_1 \\ \dot{DBR}_2 \\ \dot{DBR}_3 \end{bmatrix}^{i,j} = \begin{bmatrix} e_1^i \\ e_2^i \\ e_3^i \end{bmatrix}^T \cdot {}_{right_elm}{}^G\dot{r}_{i,j} j = 2, ..., N_{seg_elm} \tag{52} $$

where

$$ {}_{left_elm}{}^G\dot{r}_{i,j} = {}_{tire}{}^G\dot{r}_{i,j} - {}_{tire}{}^G\dot{r}_{i,j+1} \tag{53} $$

$$ {}_{right_elm}{}^G\dot{r}_{i,j} = {}_{tire}{}^G\dot{r}_{i,j} - {}_{tire}{}^G\dot{r}_{i,j-1} \tag{54} $$

Velocity of a point located at the projection of the jth belt element in the ith belt segment on the rim circumference, and described in the ith belt segment local reference frame is:

$$ {}^W_{rim_circ}v_{i,j} = \begin{bmatrix} e_1^i \\ e_2^i \\ e_3^i \end{bmatrix}^T \cdot {}^G_{rim_circ}v_{i,j} \tag{55} $$

$$ {}^G_{rim_circ}v_{i,j} = v_{rim} + \tilde{\Omega}_{rim} \times {}_{cwo}\hat{r}_{i,j} \tag{56} $$

4.10. Tire model kinematics

The components of the left sidewall force vector between i^{th} belt segment and the rim are calculated as:

$$ \begin{bmatrix} FL_1 \\ FL_2 \\ Fl_3 \end{bmatrix}^i = \begin{bmatrix} e_1^i \\ e_2^i \\ e_3^i \end{bmatrix}^T \cdot \begin{bmatrix} -k_{t1} \cdot \left(DT_1^{i,N_{belt_elm}}\right) - c_{t1}\left(\dot{DT}_1^{i,N_{belt_elm}} - \left({}^W_{rim_circ}v_{i,N_{belt_elm}}\right)_1\right) \\ -k_{t2} \cdot \left(DT_2^{i,N_{belt_elm}}\right) - c_{t2}\left(\dot{DT}_2^{i,N_{belt_elm}} - \left({}^W_{rim_circ}v_{i,N_{belt_elm}}\right)_2\right) \\ -k_n \cdot \left(DT_3^{i,N_{belt_elm}} - h_{carcass}\right) - c_n\left(\dot{DT}_3^{i,N_{belt_elm}} - \left({}^W_{rim_circ}v_{i,N_{belt_elm}}\right)_3\right) \end{bmatrix} \tag{57} $$

where $h_{carcass}$ is the tire carcass height, k_{t1} is tire sidewall tangential stiffness, c_{t1} is tire sidewall tangential damping, k_{t2} is tire sidewall lateral stiffness, c_{t2} is tire sidewall lateral damping, k_n is tire sidewall radial stiffness, and c_n is tire sidewall radial damping. The resultant force that is applied to the ith belt segment by the rim from the left sidewall is given as

$$ {}^W_{sidewall}FL^{i,j} = FL_1^i + FL_2^i + FL_3^i \tag{58} $$

The components of the right sidewall force vector between i^{th} belt segment and the rim are derived by changing the N_{belt_elm} in Eq. (57) to one (for the first belt element in right sidewall)

Similarly, the resultant force that is applied to the ith belt segment by the rim from the right the sidewall is given as:

$$ {}^W_{sidewall}FR^{i,j} = FR_1^i + FR_2^i + FR_3^i \tag{59} $$

Next, the force components within the belt segment that are generated between the neighboring belt elements are calculated. The forces exerted to the jth belt element in the ith belt segment by its neighboring elements are:

$$ \begin{bmatrix} FB_1 \\ FB_2 \\ FB_3 \end{bmatrix}^{i,j} = \left(e^i\right)^T \cdot \left[K_{belt} \cdot \left(DBR^{i,j} + DBL^{i,j}\right) - C_{belt} \cdot \left(\dot{DBR}^{i,j} + \dot{DBL}^{i,j} - {}_{rightt_elm}{}^G\dot{r}_{i,j} - {}_{left_elm}{}^G\dot{r}_{i,j}\right)\right] \tag{60} $$

where K_{belt} is the tire belt stiffness matrix and is defined as:

$$ K_{belt} = \begin{bmatrix} k_{bt1} & 0 & 0 \\ 0 & k_{bt2} & 0 \\ 0 & 0 & k_n \end{bmatrix} \tag{61} $$

In the tire belt mass matrix, k_{bt1} is the tire belt inner-tangential stiffness, k_{bt2} is the tire belt inner-lateral stiffness, and k_{bn} is the tire belt inner-radial stiffness. Moreover, C_{belt} is the tire belt damping matrix, which is defined as:

$$ C_{belt} = \begin{bmatrix} c_{bt1} & 0 & 0 \\ 0 & c_{bt2} & 0 \\ 0 & 0 & c_n \end{bmatrix} \tag{62} $$

In the belt damping matrix, c_{bt1} is belt inner-tangential damping, c_{bt2} is belt inner-lateral damping, and c_{bn} is belt inner-radial

damping. The total force vector that exerted to the jth belt element in the ith belt segment from its neighbor elements can be identified as:

$$^{W}FB^{i,j} = FB_1^{i,j} + FB_2^{i,j} + FB_3^{i,j} \tag{63}$$

Therefore, the total structural forces on the jth belt element in the ith belt segment forces are given as:

$$_{structure}^{W}F^{i,j} = {}^{W}FB^{i,j} + {}_{sidewall}^{W}FL^{i,j} + {}_{sidewall}^{W}FR^{i,j} \tag{64}$$

The torque from the tire i^{th} belt segment to the rim is written as:

$$_{belt}^{W}T^{i} = {}_{cwo}\widetilde{r}^{i,1} \times {}_{structure}^{W}F^{i,1} + {}_{cwo}\widetilde{r}^{i,N_{belt_elm}} \times {}_{structure}^{W}F^{i,N_{belt_elm}} \tag{65}$$

where

$$\widetilde{e}_2^{i} = \begin{bmatrix} 0 & -{}_{cwo}\widetilde{r}_3^{i,j} & -{}_{cwo}\widetilde{r}_2^{i,j} \\ -{}_{cwo}\widetilde{r}_3^{i,j} & 0 & -{}_{cwo}\widetilde{r}_1^{i,j} \\ -{}_{cwo}\widetilde{r}_2^{i,j} & -{}_{cwo}\widetilde{r}_1^{i,j} & 0 \end{bmatrix} \tag{66}$$

After calculating the individual force components applied to each lumped mass, the equations of motion can be written as:

$$\begin{bmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{bmatrix} = \frac{1}{m}\left\{ {}_{structural}F + {}_{internal}F + {}_{external}F + {}_{gravity}F \right\} \tag{67}$$

$$_{lumped_mass}^{G}q^{i,j} = \begin{bmatrix} x & \dot{x} & y & \dot{y} & z & \dot{z} \end{bmatrix} \tag{68}$$

For simplicity, the rim is considered as a spatial rigid body. As a result, three translational and three rotational DOF are used for describing its motion:

$$m \cdot \ddot{r} = {}^{G}F = {}_{rim}^{G}F + \sum {}_{belt}F^{i} \tag{69}$$

$$J \cdot \ddot{\phi} = {}^{G}M = {}_{rim}M + \sum_{bet} M^{i} \tag{70}$$

where ${}_{rim}F$ is the applied force vector to the rim center, ${}_{belt}F^{i}$ is the structural force from the ith belt segment, ${}_{rim}M$ is the applied torque vector to the rim center, and ${}_{belt}M^{i}$ is the applied toque vector from the ith belt segment to the rim center, and is given as:

$$\sum_{bet} M^{i} = \sum \left[\left({}_{rim_circ}r - {}_{rim}r \right) \times F^{i} \right] \tag{71}$$

The rim translational dynamics is represented in the global reference frame as follows:

$${}^{G}F = m \cdot \left({}_{rim}^{G}a \right) = m \cdot \left({}_{rim}^{G}\ddot{r} \right) \tag{72}$$

Consequently, rim translational equations of motion expressed in the rim local reference frame are given as:

$${}^{W}F = {}^{W}R_G {}^{G}F = m \cdot R_G \left({}_{rim}^{B}\ddot{r} \right) + m \cdot \left({}^{W}\omega \cdot {}_{rim}^{W}\dot{r} \right) \tag{73}$$

With the following vector form:

$$^{W}\begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} = m \,{}^{W}\begin{bmatrix} \ddot{r}_x \\ \ddot{r}_y \\ \ddot{r}_z \end{bmatrix} + m \,{}^{W}\begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} \times {}^{W}\begin{bmatrix} \dot{r}_x \\ \dot{r}_y \\ \dot{r}_z \end{bmatrix} = m \,{}^{W}\begin{bmatrix} \ddot{r}_x + \left(\omega_y \dot{r}_z - \omega_z \dot{r}_y \right) \\ \ddot{r}_y + \left(\omega_x \dot{r}_z - \omega_z \dot{r}_x \right) \\ \ddot{r}_z + \left(\omega_x \dot{r}_y - \omega_y \dot{r}_x \right) \end{bmatrix} \tag{74}$$

The applied forces to the rim center consist of the axle forces and suspension forces:

$$_{rim}^{G}F = {}_{axle}^{G}F + {}_{suspension}^{G}F \tag{75}$$

Next, the rotational dynamics for the rim is studied. The applied moment to the rim from the axle, described in the rim local reference frame, is given as:

$$_{rim}^{W}M = M_x\hat{i} + M_y\hat{j} + M_z\hat{k} = I \cdot {}_{rim}^{W}\dot{\omega} + {}_{rim}^{W}\omega \times \left(I \cdot {}_{rim}^{W}\omega \right) \tag{76}$$

where $_{rim}^{W}\omega$ is the rim angular velocity vector, and I is the inertia matrix. Additionally, the components of the torque vector are described as:

$$M_x = I_{xx}\dot{\omega}_x + I_{xy}\dot{\omega}_y + I_{xz}\dot{\omega}_z - \left(I_{yy} - I_{zz}\right)\omega_y\omega_z - I_{yz}\left(\omega_z^2 - \omega_y^2\right) - \omega_x\left(\omega_z I_{xy} - \omega_y I_{xz}\right) \tag{77}$$

$$M_y = I_{yx}\dot{\omega}_x + I_{yy}\dot{\omega}_y + I_{yz}\dot{\omega}_z - \left(I_{zz} - I_{xx}\right)\omega_x\omega_z - I_{xz}\left(\omega_x^2 - \omega_z^2\right) - \omega_y\left(\omega_x I_{yz} - \omega_z I_{xy}\right) \tag{78}$$

$$M_z = I_{zx}\dot{\omega}_x + I_{zy}\dot{\omega}_y + I_{zz}\dot{\omega}_z - \left(I_{xx} - I_{yy}\right)\omega_x\omega_y - I_{xy}\left(\omega_y^2 - \omega_x^2\right) - \omega_z\left(\omega_y I_{xz} - \omega_x I_{yz}\right) \tag{79}$$

If we assume the orientation of the rim such that $I_{ij} = 0$ when $i \neq j$, the rim coordinate system becomes a principal coordinate frame and the moment vector components are identified. Consequently, the angular acceleration of the rim is written as:

$$\ddot{\alpha} = \frac{1}{I_{xx}}\left(M_x + \left(I_{yy} - I_{zz}\right)\dot{\beta}\dot{\delta} \right) \tag{80}$$

$$\ddot{\beta} = \frac{1}{I_{yy}}\left(M_y + \left(I_{zz} - I_{xx}\right)\dot{\alpha}\dot{\delta} \right) \tag{81}$$

$$\ddot{\delta} = \frac{1}{I_{zz}}\left(M_z + \left(I_{xx} - I_{yy}\right)\dot{\beta}\dot{\alpha} \right) \tag{82}$$

The equations for the sidewall moments that exist between the belt segments and the rim are expressed in the wheel local reference frame. In order to express these equations in the global reference frame, the following transformation matrix is used:

$${}_{rim}R(t) = R_{Z,\delta(t)}R_{X,\alpha(t)}R_{Z,\beta(t)} \tag{83}$$

It should be noted that the transformation matrix is a function of time, and is recalculated at every time step based on the spatial orientation of the rim. The final moment vector at the spindle resulting form the tire sidewall forces, expressed in global reference frame is calculated as:

$$_{sidewall}^{G}M^{i} = {}_{rim}R(t) \cdot {}_{sidewall}^{B}M^{i} \tag{84}$$

where $_{sidewall}^{G}M$ is the sum of individual torque vectors applied by the belt segments to the tire rim:

$$ {}^{B}_{spindle}M = \sum_{i=1}^{N_{belt_seg}} \left({}^{B}_{sidewall}M^{i} \right) \quad (85) $$

5. Tire model parameterization

Tire model parameterization is defined as the set of experiments and data processing methods that are performed to acquire the input parameters for the tire model simulation. There are different methods that can be used for tire parameterization. These methods range from completely empirical to semi-empirical methods. For the model developed in this study, a complete set of tire parameterization methods for individual parameters was proposed in another study from the authors (Sandu et al.).

6. Tire-terrain interaction

Once the tire structure and tire material properties are modeled and implemented in a mathematical framework, the interface between the model and road surface should be established. This interface searches for the nodal points that are close to the ground (contact search algorithm), and once the contact is detected, the algorithm applies the required contact condition (contact interface algorithm). The detailed description of the tire-terrain interaction model is outside the scope of this paper, and is presented in the companion paper (Sandu et al., 2019).

7. Simulation platform

In order to characterize the configuration of the test rig in the simulation environment, an application platform was designed in order to accommodate the communications with the multibody dynamics solver. The spindle carriage of the VT Terramechanics test rig (Naranjo, 2014; Jimenez and Sandu, 2015; Sandu, 2015, 2008) was represented using a quarter-car model (with high suspension stiffness and damping values), and was implemented in a separate module. The stiffness of the quarter car model corresponds to the first vibrational mode of the rig structure, and the vertical damping was included to eliminate unwanted model vibrations in the vertical direction during simulation. If the normal load on the tire is prescribed during the simulation, an implemented virtual load actuator applies additional force to control the normal load.

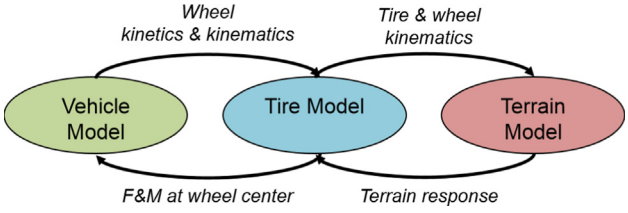


Fig. 5. The communication data flow between the tire model modules during the full vehicle simulation.

The tire model includes its own ordinary differential equations (ODE) solver. The CVODE solver from Sundials (Hindmarsh, 2005) was used, which is a variable-order, variable-step solver based on Adams-Moulton formulas. At every time step, the vehicle model, which is described in a multibody dynamics framework, provides the wheel kinetics and kinematics variables to the tire model. The time step for the tire model solver was chosen as half the time step set for the multibody dynamics solver. This is due to the fact that vehicle model requires extra calculations in the middle of its fixed time intervals to improve the accuracy. These extra calculation results are provided to the external solver for maximizing the vehicle model ODE solver performance. Next, the tire model updates the position and velocity state vectors of the tire. Using this new tire configuration, the terrain model exploits the contact conditions, which results in the tire/ground deflection and stress distribution in the contact patch. The normal and shear stress fields are feedback to the tire model, which are used for solving the tire equations of motion. At the end of this step, the tire model calculates three forces and moments at the spindle and feeds them back to the vehicle model. An overview of the discussed procedure is shown in Fig. 5. Additionally, as for the tire-vehicle interface, the data communication routines are developed such that they follow the standard formats from Standard Tire Interface (STI) practices (Oosten, 1997; Van Oosten, 1999).

It should be noted that great attention was given to the optimization of the tire model performance in order to make it a practical option for full vehicle simulations. This has become possible by applying some parallel and multi-processing techniques to the architecture of the program. In this regard, the tire model stiffness matrix is partitioned into smaller matrices corresponding to tire belt element nodes located in multiple rings across tire width. At each simulation time step, the task of calculating forces on belt elements and solving the equations of motion for each ring is assigned to individual CPUs. The computation of each local matrix is totally independent and so parallelization of these computations is straightforward and can be naturally explored.

To evaluate the performance of the multi-processing program architecture, the free rolling tire on a non-deformable terrain was simulated with and without utilizing the multi-processing feature. The total simulation time was set to 10 s and 1800 degrees of freedom were used in the model. The solver time step was 0.001 s. The computational run time for three main simulation tasks were reported in Fig. 6. It can be seen that by using the multi-processing techniques, a great improvement in the time required for assem-

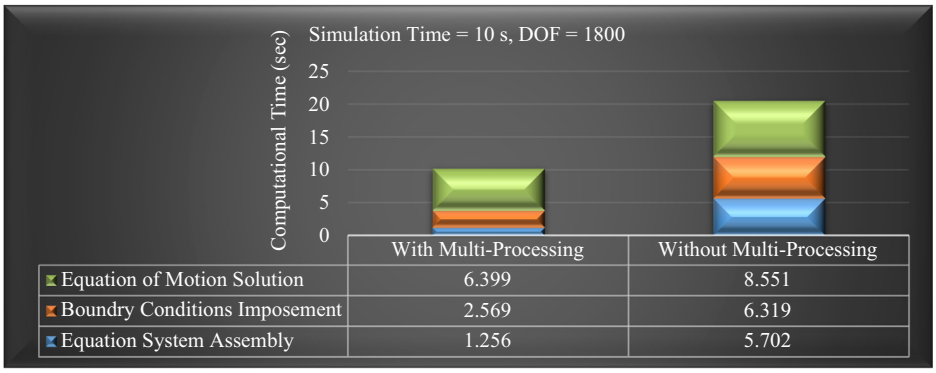


Fig. 6. Contribution of each task group in the total processing time for simulations with and without utilizing multi-processing architecture.

bling the model equation system and imposing the boundary conditions is observed. With the multi-processing techniques used, the computational time for performing 10 s simulations is close to real-time (10 s total).

8. Conclusions

In order to model the dynamic behavior of the tire on soft soil, a three-dimensional discretized approach was used to describe the tire structure characteristics. The main objective of this method was to obtain an off-road tire model with soft-soil capabilities and an accurate enveloping behavior of the over (off-)road perturbations. This paper is the first of three. Together, the three papers completely describe the tire modeling approach, tire-terrain interaction, tire parameterization, and model validation.

In this first paper out of three we address the modeling of the tire structure. Kelvin-Voigt elements were used as the main force elements between tire/rim elements due to their accuracy in expressing tire visco-elastic behavior, and their efficiency compare to more complex models. The tire belt was discretized into multiple belt segments that are suspended on the rim by springs and dampers. Belt segments were further discretized into belt elements to account for the lateral road inputs. The required discretization level was identified as a function of minimum road wavelength and computation load on the solver.

The developed model, named Hybrid Soft Soil Tire Model (HSSTM) was integrated with the vehicle model CarSim to simulate vehicle performance on deformable terrains. To optimize the computational time of the code, different techniques were used in memory allocation, matrix manipulation, and multi-processing. The performance gain from these treatments was discussed.

The HSSTM tire model uses empirical equations to describe the stress and strain behavior of the non-deformable and deformable terrains. The implementation of the underlying empirical equations, and discussion over the terrain model parameters and performance results are given in the companion paper Part II (Sandu et al., 2019).

The tire parameterization procedures were conducted using a variety of procedures that are handled in a separate paper, Part III (Sandu et al.). Simulation results and experimental data are presented in this part, too, as the model validation is discussed.

9. Disclaimer

Reference herein to any specific commercial company, product, process, or service by trade name, trademark, manufacturer, or otherwise, does not necessarily constitute or imply endorsement, recommendation, or favoring by the United States Government or the Department of the Army (DoA). The opinions of the author expressed herein do not necessarily state or reflect those of the United States Government or the DoA, and shall not be used for advertising or product endorsement purposes.

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